

An Occupational Taxonomy by Professional Characteristics: Implications for Research

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Theorists concerned with the nature of professionalism and researchers investigating professionals in the work environment often assume that professional occupations are a unique occupational type. To test this assumption, an occupational taxonomy based on nine dimensions of professionalism was derived empirically. The taxonomy represented expectations only to a moderate degree. Several implications were noted for the use of occupational taxonomies in the generalization of research findings on one occupational group to other occupational groups in the same classification.

In studying the behavior and/or attitudes of professionals in organizations, most researchers make two implicit assumptions. The first assumption is that professional occupations are readily identifiable. The second assumption is that professional occupations have essentially similar characteristics. Both assumptions are necessary if researchers hope to generalize findings for one occupation to a group of occupations deemed professional. Unfortunately, most researchers accept available categorizations of professional occupations (such as those of the U.S. Bureau of the Census) a priori with little concern for the heterogenous nature of such classifications. Even though there are many occupational taxonomies, these have not been derived purposefully from representative theories of professionalism. Thus, for example, the occupations in the occupational category of professional and technical occupations of the Census Bureau have been shown to vary widely on professional dimensions (Cullen, 1978).

Although the arbitrary selection of professional occupations probably lacks utility, occupational categorizations are useful. Just as organizational theorists need organizational classifications for scientific generalizations to specific organizational types (McKelvey, 1975; Pinder & Moore, 1979; Pinto & Pinder, 1972), researchers investigating organizational behavior need typologies of occupational groups for scientific generalizations pertinent to the behavior and/or attitudes of individuals in similar occupational

groupings. Following Hempel (1965, p. 146), it has been argued by McKelvey (1975, p. 510) that: "The classificatory type concept is the most useful one for the scientific activity of inducing theories about certain phenomena and deducing testable hypotheses. It provides the basis for explanation, prediction, and scientific understanding by identifying uniformities in the phenomena about which laws and principles may be formed. . . ."

Various systems of occupational classification not necessarily intended to identify professional or nonprofessional occupations have been used for practical and/or scientific generalizations. Personnel psychologists have recognized that the criterion validation of instruments designed to measure job qualifications or job performance can be generalized from one job to other jobs of a similar type (Arvey & Mossholder, 1977; Mobley & Ramsay, 1973). Taxonomies of work emphasizing interests or abilities (e.g., Kuder, 1951) have been found useful for differentiating occupations into groups relevant for career guidance and personnel selection (Dawis & Lofquist, 1975). Similarly, wage and salary administration systems as well as performance appraisal systems need taxonomies of similar jobs (at least within the organization) to identify groups of occupations to be managed similarly (Cornelius, Carron, & Collins, 1979).

Although many occupational taxonomies provide viable classification systems for researchers concerned with work and organizational behavior, it remains unknown whether occupations can be clustered homogeneously based on occupational variables derived from theoretical conceptualizations of professional dimensions. Some theorists have hypothesized explicitly that professions are unique (e.g., Goode, 1969) and, thus, are a readily identifiable and similar occupational grouping. Likewise, authors who use the word "profession(als)" in the titles of their behavioral science articles implicitly assume that their research pertains to an accepted occupational type. To test these assumptions, this paper presents an occupational classification system derived empirically and based on nine of the many dimensions of professionalism identified by the theoretical literature.

PROFESSIONALISM

Although identifications of the attributes of a "true" profession abound (e.g., Greenwood, 1957), any author's particular list is somewhat arbitrary in both the selection of elements and in the magnitude of an element (such as education) deemed necessary for professional status. There is, however, some consensus among authors, as can be seen in Table 1, a summary of classical and often-referenced statements on the nature of professionalism.

TABLE 1
Selected Dimensions of Professionalism by Citing Authors

Dimension	Author														
	Akers (1970)	Barber (1963)	Caplow (1954)	Carr- Saunders & Wilson (1933)	Cogan (1953)	Flexner (1915)	Foote (1953)	Goode (1957; 1969)	Green- wood (1957)	Gross (1958)	Lewis & Maude (1953)	Moore (1970)	Pavalko (1971)	Vollmer & Mills (1966)	Wilensky (1964)
Complex occupation	+	+		+	+	+	+	+	+	+		+	+	+	+
Self-employed				+							+				
Complex relationships with people				+	+	+		+							
Altruistic															
service					+	+		+				+			
Long training	+	+		+		+		+	+		+	+	+		+
Organized	+	+	+	+		+	+	+	+	+	+	+	+	+	+
Code of ethics		+	+	+				+	+		+	+	+		+
Competence tested				+							+	+			
Licensed	+		+					+							+
High income				+				+	+						
High prestige		+						+							

Note. Aspects of this table were adapted from Millerson (1964, p. 5).

METHOD

Occupations

The occupation was the unit of analysis. Two hundred sixty-seven nonfarm occupations were selected from the detailed occupational classifications of the U.S. Bureau of the Census (1973, pp. 1–11) and the U.S. Department of Labor (1974, pp. 853–876). The nature of these classification systems precluded dealing with all possible occupations. The objective in selecting this nonrandom sample was to eliminate internal occupational specializations as well as broad occupational classifications that contained many different occupations. The initial sample of occupations covered approximately 55% of the *individuals* in the 1969 experienced civilian labor force. Missing data for the variables used in this study reduced the sample to 203 occupations.

Measures

Operational indicators for 9 of the 11 dimensions identified in Table 1 are shown in Table 2 beside their respective theoretical concepts. Measures of ethical code existence or use and altruistic service were not included in the major cluster analysis. Some data on ethical codes and the service goals of the national organization of the occupation were available from a survey of occupational associations (Cullen, 1978). However, because the use of these data would have severely reduced the sample size to barely over 100 occupations, it was decided to eliminate these two variables and maximize the number of occupations in the taxonomy reported below in Table 3.

Data for the taxonomy were gathered from published sources, with the *Dictionary of Occupational Titles* (DOT) (U.S. Department of Labor, 1965) providing information for most variables. The DOT is considered a general source of occupational information in social and behavioral science research (Cain & Treiman, 1981; Spenner, 1980). Not only is the DOT based on a functional taxonomy of work (Fine & Heinz, 1958) but it has also provided variables for several other occupational taxonomies (e.g., Dawis & Lofquist, 1975). DOT information has previously been merged with various Census classifications (e.g., Broom, Jones, Jones, & McDonnell, 1977; Cullen & Novick, 1979). Additional information on selection of the occupational sample, measurement of variables, and merging the DOT and Census Bureau occupational classifications was reported in Cullen (1978).

Statistical Procedure

Following an algorithm developed by Johnson (1967) and using the nine empirical indicators of the professional dimensions described in Table 2, a hierarchical cluster analysis was performed on 203 occupations.

TABLE 2
Selected Dimensions of Professionalism by Corresponding Empirical Indicators

Theoretical dimension	Empirical indicator
Comples relationships with people	People complexity: A 9-point scale noting the involvement and complexity of an occupation's tasks as they are applied to people. Source: U.S. Department of Labor (1965). ($M = 1.30$, $SD = 2.04$)
Complex occupation	Data complexity: A 9-point scale noting the involvement and complexity of an occupation's tasks as they are applied to data. Source: U.S. Department of Labor (1965). ($M = 4.71$, $SD = 2.70$)
Self-employed	Self-employment: The number of self-employed occupational group members divided by the total number of individuals in the occupational group. Source: U.S. Bureau of the Census (1973, Tables 28, 43). ($M = .10$, $SD = .17$)
Organized	Organized: A dummy variable noting the existence of a major national occupational association. Sources: Fisk (1974); U.S. Department of Labor (1971). (78.7%)
Competence tested	Exam utilization: A 3-point scale where 0 = no examination necessary to enter the occupational group; 1 = passing examination improves employment probability; and 2 = passing examination necessary to enter the occupational group. Source: U.S. Department of Labor (1972-3). ($M = .41$, $SD = .71$)
Licensed (prevalence of licensure)	Licensure: The number of states in which an occupation is licensed, certified, or registered by the state government. Sources: Council of State Governments (1968); U.S. Department of Labor (1974). ($M = 8.86$, $SD = 19.10$)
Long training (length of training)	Education: The median education of the occupational group's experienced civilian labor force. Source: U.S. Bureau of the Census (1973, Table 1). ($M = 13.19$, $SD = 2.68$)
High income	Earnings: The mean earnings of the occupational group's experienced civilian labor force. Source: U.S. Bureau of the Census (1973, Table 19). ($M = 9085$, $SD = 3989$)
High prestige (prestige rating)	Prestige: An occupation's prestige ranking. Source: Treiman (1975). ($M = 44.94$, $SD = 15.03$)

Note. Means and standard deviations are for the 203 occupations used in this study.

Hierachical cluster analysis has been suggested as an appropriate technique for identifying job similarities (Mobley & Ramsay, 1973).

Because variables such as earnings, education, and data complexity were not measured on the same scale, it was necessary to standardize all measures included in the analysis. The use of Z scores avoided weighting the distances between observations in favor of "large" variables such as earnings (see Hartigan, 1975, pp. 59-60 for further explanation).

TABLE 3
Occupations by Clusters

Cluster 1		
Chiropractor	Optometrist	Podiatrist
Dentist	Osteopathic physician	Veterinarian
Lawyer	Physician	
Cluster 2		
Accountant	Computer programmer	Mechanical engineer
Actor	Computer system analyst	Meteorologist
Advertising agent	Credit manager	Mining engineer
Aeronautical engineer	Dancer	Musician
Apparel store manager	Dietician	Newspaper reporter
Architect	Draftsman	Occupational therapist
Astronomer	Economist	Oceanographer
Author	Electrical engineer	Personnel worker
Bank officer	Elementary/secondary teacher	Pharmacist
Biochemist	Geophysicist	Psychologist
Broadcast technician	Historian	Public relations worker
Buyer	Industrial designer	Railroad station agent
Chemical engineer	Industrial engineer	Real estate agent
Cehmical technician	Judge	Sales engineer
Civil engineer	Landscape architect	School counselor
Clergyman	Mathematician	Singer
College/university teacher	Mathematician technician	Sociologist
College administrator		Statistician
		Tool programmer
Cluster 3		
Aircraft mechanic	Garbage collector	Plasterer
Asbestos & insulation worker	Gardener	Police officer
Automobile body repairman	Gasoline station attendant	Postal clerk
Automobile mechanic	Gasoline station manager	Postmaster
Baker	Glazier	Power station operator
Bank teller	Guard	Power truck operator
Barber	Hairdresser	Pressman
Bartender	Housekeeper	Proofreader
Bellman	Industrial engineering technician	Radio & television repairman
Boilermaker	Insurance salesman	Radio operator
Bookbinder	Janitor	Railroad brakeman
Bookkeeper	Keypunch operator	Railroad conductor
Bootblack	Laundress	Railroad switchman
Bricklayer	Legal secretary	Real estate salesman
Busdriver	Librarian	Receptionist
Butcher	Library assistant	Recreation attendant
Cabinetmaker	Licensed practical nurse	Registered nurse
Calculating machine operator	Locomotive engineer	Roofer
Carpenter	Locomotive fireman	Sales Clerk
Carpenter's helper	Logger	Seaman
Cashier	Longshoreman	Seamstress
	Loom fixer	Secretary
	Machinist	Sheet metal worker

TABLE 3 (Continued)
Occupations by Clusters

Cluster 3		
Cement mason	Mail carrier	Shipping clerk
Child care worker	Mechanical engineering technician	Shoe repairman
Claims adjuster	Medical laboratory technologist	Solderer
Claims examiner	Medical record librarian	Statistical clerk
Coal miner	Medical secretary	Stenographer
Computer operator	Messenger	Stone cutter
Cook	Metal molder	Surgical technical
Demonstrator	Meter reader	Surveying assistant
Dental assistant	Milliner	Surveyor
Diesel mechanic	Millwright	Tabulating machine operator
Dishwasher	Motion picture projectionist	Tailor
Electric power lineman	Newsboy	Taxi driver
Electrotypist/stereotypist	Occupational therapy assistant	Telephone cable splicer
Elevator operator	Operating engineer	Telephone installer/repairman
Embalmer	Optician	Telephone lineman
Engraver	Painter	Telephone operator
Farm products buyer	Paper hanger	Tile setter
File clerk	Parking attendant	Tool & die maker
Firefighter	Pattern model maker	Truck driver
Flight attendant	Physical therapist	Typist
Flight engineer	Physical therapy assistant	Upholsterer
Food store manager	Piano tuner	Usher
Funeral director		Waiter/waitress
Furniture & wood finisher		Welder
Furrier		

RESULTS

Occupations seemed to group into three clusters. The three occupational types suggested by the cluster analysis are shown in Table 3. The means for professional variables within each cluster are shown in Table 4.

A maximum of three clusters were identified based on the ratio $ND's$ within clusters/ $ND's$ within and between clusters $\leq D$ max, where D is the distance between any two observations, $ND's$ are the number of distances, and D max is the *maximum* distance found between any two observations *within* a cluster. As this ratio approaches 1, proportionately more observations are closer to other observations within their respective clusters than they are to observations in other clusters. The ratio for three clusters was .58, and the ratio for four clusters was .48.

To some degree, the clusters identified from this sample of occupations represent conventional views on the nature of professionalism. The small group of predominantly medical occupations shown in Cluster 1 are probably the pinnacle of professional status, being most professionalized on all attributes except data complexity. Cluster 3, the largest of the

TABLE 4
Means of Professional Variables by Clusters

Variable	Total (all clusters)	Cluster		
		1	2	3
People complexity (0-8)	1.3	7.2	1.7	0.7
Data complexity (0-8)	4.7	7.1	7.3	3.6
Self-Employment (%)	10.2	74.7	0.1	0.1
Organized (% scored 1)	78.7	100.0	89.3	75.7
Exam utilization (0-2)	.4	2.0	0.3	0.4
Licensure (No. states)	8.9	49.8	14.6	4.5
Education (years)	13.2	19.4	16.0	11.8
Earnings (1969 \$)	9,085.9	21,712.2	12,068.7	7,214.9
Prestige (0-100)	44.9	66.2	59.5	36.8

occupational groupings, was clearly the least professionalized, with low means on nearly all the professional dimensions. Cluster 2 contained most of the accepted professions that are employed by formal organizations, thus differing from Cluster 1 to a large degree based on the extent of self-employment. Cluster 2 also had considerably less licensure and examination utilization than did Cluster 1.

Although three clusters were selected as the maximum number of groupings that maintained a reasonable ratio of *ND*'s within clusters/total distances within and between clusters, it was possible to consider this sample of occupations as forming only two clusters. When Clusters 1 and 2 were merged, a ratio of .59 resulted. Essentially, a classification of professional versus nonprofessional emerged.

To examine the possibility that some occupations were misclassified because only 9 of the 11 dimensions of professionalism were used in the taxonomy (altruistic service and the ethical code were not included), an additional analysis of a limited sample of occupations ($n = 123$) was conducted. Data from Cullen's (1978) study, based in part on a survey of occupational associations, provided a dummy variable on the existence of an ethical code and the occupational association's full-time executive officer's estimate (from "extremely important" to "not at all important") of the association goal of "encouraging service for the betterment of

society'' (p. 249). Of the 123 occupations examined, and with the two additional variables included in the cluster analysis, all occupations but one (claim examiners moved from Cluster 2 to 3) were classified into the groups reported in Table 3. Thus, these data suggest that even with more dimensions of professionalism included in the cluster analysis, an essentially similar taxonomy would be produced.

DISCUSSION

Theorists concerned with the nature of professionalism and researchers concerned with professionals in the work environment are not entirely wrong in assuming that the professions form a unique occupational type. The data examined in this study, however, suggest caution in the generalization of research based on such assumptions. While it has been shown that classes of occupations can be identified by professional elements, a reasonable proportion of occupations were found to have attributes closer to occupations in other clusters than to occupations in their own cluster. This occurred regardless of whether the occupational sample was viewed as dichotomy (professional/nonprofessional) or nonprofessions versus two types of professions.

Also of note was the finding that some occupations usually described by researchers as professional were missing from their expected classification. Registered nurses, librarians, opticians, and physical therapists might all be considered accepted professions. A more detailed examination of these occupational groups showed that they had, on average, lower earnings than did the occupational groups in Clusters 1 and 2 while, at the same time, they lacked sufficient development on other professional dimensions to offset their lower earnings.

In terms of the derivation of hypotheses and the generalization of results, there are several recommendations for behavioral science researchers studying occupational groups typically considered "professional." First, give most weight to previous studies dealing with individuals from the same occupation. Second, derive hypotheses and base generalizations on findings for occupations classified in taxonomies more specific than those identifying professional occupations. For example, occupational taxonomies based on task variables may more appropriately identify similar occupations for studies of task design and job satisfaction. Taxonomies based on education, earnings, and prestige may do the same for labor relations research. Third, because professionalized occupations are somewhat similar, previous findings from research on one type of professional group (e.g., physicians) are useful for hypotheses pertaining to other groups; but differences between professional occupations should be anticipated.

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Received: February 10, 1982.